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GENERATIVE AI-BASED SYNTHESIS AND SUPER-RESOLUTION OF BRAIN FMRI IMAGES

SUPERVISOR

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Kelt: Budapest, 2025.11.30

Sajben Dániel

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Dániel Sajben

STUDENT DECLARATION (TRANSLATION)

I, **Dániel Sajben**, the undersigned, hereby declare that the present BSc thesis work has been prepared by myself and without any unauthorized help or assistance. Only the specified sources (references, tools, etc.) were used. All parts taken from other sources word by word, or after rephrasing but with identical meaning, were unambiguously identified with explicit reference to the sources utilized.

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Budapest, 30 November 2025

Sajben Dániel

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Dániel Sajben

Hallgatói nyilatkozat generatív mesterséges intelligencia alkalmazásáról

☐ **Nem használtam** semmilyen generatív MI segédeszközt.

☒ **Használtam** generatív MI segédeszközt. Az MI-vel generált tartalmakat ellenőriztem, a generált kimenetek valóságtartalmáról meggyőződtem, az alábbi táblázatban megfelelően jelöltem minden használatot.

Felhasználási módok	**Generatív MI eszköz(ök) neve**	**Érintett részek** (fejezet, oldalszám, hivatkozás)	**Használat becsült aránya** (felhasználási módonként)
Irodalomkutatás	ChatGPT 4o/5 (Deep Research), Gemini 2.5/3 Pro	Identification of relevant research papers, which were subsequently read and cross-checked. Paragraph 6 References, pages 50-52 No specific thesis section is directly attributable to AI output.	30 %
Prompt lényegi része	"Find relevant papers on diffusion models, VQ-GANs, VQ-VAEs, autoencoders, super-resolution, medical imaging, evaluation metrics, etc. / Check if all the citations are correctly formatted."		
Magyarázat	Identification of papers was AI-assisted, but reading and verification were manual. The AI contribution is estimated at 30% of the *search process*, not the content creation. Around half of the papers used were suggested by AI, but since I read all of them a ~30% help could be a rough estimate, for the time saved.		
Programkód generálása	Claude Sonnet 4.5, Gemini 2.5/3 Pro, ChatGPT 4o/5	All Python files contain AI-generated code suggestions through Visual Studio Code's built-in AI-assisted autocompletion features.	35 %
Prompt lényegi része	"Debug this code / Why is there a dimensional mismatch? / Provide a template for implementing a diffusion or autoencoder architecture in Python/PyTorch. / How could I optimize this training"		
Magyarázat	Out of the ~8500 lines of code around ~3000 contains either AI assistance, Autocompletion or polishing/optimalization. ~35% The core logic for the VQ-GAN and Diffusion architecture was implemented manually or adapted from papers.		

Új ötletek, megoldási javaslatok generálása	Gemini 2.5/3 Pro	Guidance on designing a VQ-based architecture for the discrete latent space. Section 2.4	20 %
Prompt lényegi része	“How can I make my diffusion model more stable or reduce latent-space variance?”		
Vázlat létrehozása (szövegstruktúra, vázlatpontok)	ChatGPT 4o/5, Gemini 2.5/3 Pro	The whole thesis uses paragraph structuring support without copying any entirely AI-generated text.	20%
Prompt lényegi része	“How would you restructure this paragraph to align with a research-style thesis?”		
Magyarázat	Since no paragraph or structure was entirely taken or copied just insights for polishing purposes a ~20% could be a fair estimate.		
Szövegblokkok létrehozása	ChatGPT 4o/5, Gemini 2.5/3 Pro	No verbatim copied paragraphs, AI was used only for polishing, terminology suggestions, and improving clarity throughout the whole thesis. Abstract and declarations were heavily based on the suggestions (especially the translation) pages 10-11.	20 %
Prompt lényegi része	“Could you refine this paragraph? / What is a better scientific term for this? / How would you translate this expression?” Only structural and linguistic improvements were used, not full paragraph copies.		
Magyarázat	Since no paragraph was entirely taken or copied just phrases and structural insights for polishing purposes and translation a ~20% could be a fair estimate.		
Képek generálása illusztrációs célból	0	0	0 %
Prompt lényegi része	-		
Adatvizualizáció, grafikonok generálása adatpontok alapján	Claude Sonnet 4.5	PCA plotting code generated to support a single illustrative figure. In Section 4.4.1 Figure 4.1.	100%
Prompt lényegi része	„Generate a matplotlib PCA ploting python code of the subject distribution.”		
Magyarázat	The code was used solely to produce one supporting visualization, but 100% of the plotting is AI generated since I only had one plot for data visualization.		
Prezentáció készítése	0	0	0 %
Prompt lényegi része	-		
Egyéb (Paper summarization)	Google NoteBookLM	AI-generated summaries used solely to gain an initial overview of long research papers.	100 %
Prompt lényegi része	Google NotebookLM automatically summarized long papers to accelerate literature understanding; no specific prompts were required.		

Magyarázat	Since all the papers were summarized before reading, 100% of the cases this tool was used.
Összesített százalékos érték (a feladat érdemi részére nézve)	~ 30 %
Összesített érték rövid, szöveges indoklása: I used generative AI tools primarily for self-education, literature exploration, linguistic refinement, and workflow acceleration. All referenced research papers were independently verified and fully read before use. AI-generated content was not used to replace original research contributions but to support tasks such as code autocompletion, debugging assistance, template generation, documentation (such as readme or comments), formatting, paragraph polishing, and summarization of long texts. Critical research decisions, including model design, architectural choices, hyperparameter selection, and methodological development, were made independently, based on my own analysis and understanding. Furthermore, the interpretation of experimental results and the formulation of the final scientific conclusions remain exclusively my own intellectual work, uninfluenced by generative outputs.	

Figyelem: a fenti százalékértékektől függetlenül az MI-vel generált minden tartalomért a szerző a felelős!

Sajben Dániel

Abstract (Hungarian Translation)

A mesterséges intelligencia gyors fejlődése az orvosi képalkotás területén új lehetőségeket teremt a diagnózisok felgyorsítására és az emberi hibák csökkentésére. Ugyanakkor a funkcionális mágneses rezonancia képalkotás (fMRI) adatainak előállítása költséges és időigényes, ami korlátozza a nagy, robusztus AI-modellek tanításához szükséges adatállományok elérhetőségét. Ez a dolgozat az adatok korlátozott mennyiségéből és minőségéből fakadó kihívást egy generatív AI-alapú folyamat fejlesztésével próbálja megtoldani, amely az fMRI képszintézist és a „szuperfelbontást" (super-resolution) célozza a diagnosztikai hasznosság növelése érdekében.

Egy kétlépcsős, látens diffúziós modell (LDM) keretrendszer került kidolgozásra: először egy 3D vektorkvantált generatív adverzariális hálózat (VQ-GAN) tömöríti az fMRI térfogatok adatait egy diszkrét, strukturált látens térbe, stabilitást és hűséget biztosítva. Másodszor egy feltételes 3D diffúziós modell javítja a kis felbontású bemeneteket, amelyet a classifier-free guidance (CFG) irányít a jobb strukturális pontosság érdekében. Kiterjedt patch-alapú előfeldolgozási és „csúszó-ablakos" (sliding-window) generálás technikák kerültek alkalmazásra a GPU-memória korlátainak leküzdésére és a teljes agy rekonstruálásának lehetővé tételére.

A kísérleti eredmények azt mutatják, hogy a CFG-alapú finomhangolás jelentősen szűkíti a különbséget a szuperfelbontású és az eredeti nagy felbontású felvételek minősége között, a PSNR és SSIM metrikák tekintetében megközelítve a hagyományos interpolációs módszerek szintjét, miközben megőrzi a finom anatómiai részleteket. Az adverzariális finomhangolás tovább javítja az észlelt élességet, élethű rekonstrukciókat eredményezve lerontott bemenetekből.

Ez a munka megmutatja, hogy a diffúzió-alapú szuperfelbontás hatékonyabbá teheti az fMRI képalkotást a vizsgálati idő csökkentésével és az archív adatok minőségének javításával, praktikus és skálázható megoldást kínálva az orvosi képalkotási munkafolyamatok számára. A teljes kódbázis és a betanított modellek nyilvánosan elérhetők a jövőbeli kutatási és klinikai alkalmazások támogatására.

Abstract

The rapid advancement of AI in medical imaging has created opportunities to accelerate diagnosis and reduce human error. However, functional magnetic resonance imaging (fMRI) data are costly and time-consuming to acquire, limiting the availability of large datasets needed to train robust AI models. This thesis addresses the challenge of limited data quality and quantity by developing a generative AI pipeline for fMRI image synthesis and super-resolution to improve diagnostic utility.

A two-stage latent diffusion model (LDM) framework was proposed: first, a 3D vector-quantized generative adversarial network (VQ-GAN) compresses fMRI volumes into a discrete, structured latent space, providing stability and fidelity. Second, a conditional 3D diffusion model enhances low-resolution inputs, guided by classifier-free guidance (CFG) for improved structural accuracy. Extensive patch-based preprocessing and sliding-window inference strategies were implemented to overcome GPU memory limitations and enable full-brain reconstruction.

Experimental results demonstrate that CFG fine-tuning significantly narrows the quality gap between super-resolved and original high-resolution scans, achieving PSNR and SSIM metrics close to traditional interpolation baselines while preserving fine anatomical detail. Adversarial fine-tuning further improves perceptual sharpness, producing realistic reconstructions from degraded inputs.

This work shows that diffusion-based super-resolution can make fMRI imaging more efficient by reducing scan times and enhancing archival data quality, offering a practical and scalable solution for medical imaging workflows. The codebase and trained models are publicly available for future research and clinical adaptation.

1 Introduction

The primary motivation for this research is the rapidly evolving field of AI-assisted medical imaging. AI models can significantly accelerate the evaluation of various medical scans and help prevent human errors during assessment. However, these models require vast amounts of training data, and functional magnetic resonance imaging (fMRI) images are particularly expensive to produce. Consequently, there is often a scarcity of data to train efficient and robust models for diagnostic tasks. This challenge was the initially motivation behind my research to develop a generative AI solution using diffusion technologies to create synthetic fMRI images, which could then be used to train other evaluation models. That realization, stemming from the initial experimentation with generative technology, indicated that the project's scope could be broadened to super-resolution.

Super-resolution is a technique that allows us to increase the resolution of low-quality fMRI images. This capability could help reduce scan times in clinical settings and improve the usefulness of previously collected, lower-resolution scans. Nevertheless, it would also make it possible to generate high-quality fMRI images for accurate diagnosis in regions that cannot afford high-end scanners. Even in Hungary, the most widely used machines are 1.5-tesla systems, which limit the ability to diagnose certain diseases. If these images could be artificially enhanced to 3-tesla quality, they would become far more useful for diagnostic tasks. Furthermore, some individuals have implants that prevent them from entering a 3-tesla or higher electromagnetic field but allow them to safely undergo imaging at 1.5 tesla. This technology would enable such patients to obtain 3-tesla-quality fMRI images without ever being exposed to a higher magnetic field, thereby improving their diagnostic outcomes.

Direct conclusions were drawn from the first part of the thesis, which focus was synthetic images generation. Those experiences were used, along with findings from research papers and public libraries, to implement a state-of-the-art super-resolution model, which is arguably even more useful than simply generating images to augment datasets.

1.1 Technical Approach and Architectural Evolution

The proposed approach focuses on efficiency while addressing the challenges of generating highly structured 3D images. Generating a single fMRI volume using a standard denoising diffusion probabilistic model (DDPM) is computationally expensive and time-consuming. To address this, my approach utilizes an autoencoder to compress the fMRI data into a compact latent space, perform the diffusion process in this lower-dimensional space, and then decode the results back into a full image. This method, often referred to as a latent diffusion model (LDM), has a higher initial cost to train the autoencoder but makes the final image generation process significantly faster and more computationally feasible.

My initial attempts encountered several key challenges. Experiments were conducted with a 2D latent space to leverage pre-made diffusion models and reduce computational load, but the resulting images were poor in quality because the critical structural details were lost. Another approach involved a 3D latent space, but its structure varied so erratically that the diffusion model was unable to learn its distribution, and failed to converge, resulting in significant structural artifacts..

Based on these observations, the methodology was refined to the super-resolution model. A 3D vector-quantized generative adversarial network (VQ-GAN) was implemented as the autoencoder to compress our data. Its "quantized" latent space provides the consistency and stability needed for a generative model to succeed. In the second stage, a 3D diffusion model is trained to enhance low-resolution inputs by denoising them into high-quality latent representations, conditioned on their compressed low-resolution counterparts. This two-stage approach yielded qualitatively excellent results.

1.2 Implementation and Availability

Multiple data handling strategies were extensively explored during the experiments, as the GPU memory was a major constraint. The final solution was to pre-process the data into small 64x64x64 chunks, which could be easily handled by our GPU. This pre-processing step prevents the CPU from becoming a bottleneck during training, as all heavy computations are done once beforehand. For inference, these individually processed patches are then stitched back together using a smooth blending technique to reconstruct the full, high-resolution brain volume. The MONAI library was utilized to easily and efficiently implement this patch-based approach.

The entire implementation, including preprocessing, training, and inference code, is publicly available on GitHub for further exploration and future development. The final models are also available to download from google drive.

GitHub:

- https://github.com/SajbenDani/Thesis

Models:

- https://drive.google.com/drive/folders/1-PazTz0ylF8VTsglEHhgy1C53LKIJ7R5?usp=sharing

2 Background and Theoretical Foundations

2.1 Functional Magnetic Resonance Imaging (fMRI)

Functional magnetic resonance imaging (fMRI) is a non-invasive neuroimaging technique that measures brain activity by detecting changes in blood oxygenation, known as the blood-oxygen-level-dependent (BOLD) contrast [1]. By tracking these BOLD signals over time, fMRI produces dynamic activation maps that reveal which brain regions are active during specific tasks, making it essential for studying cognition and neurological disorders [2].

2.2 Generative Models for Image Synthesis

2.2.1 Theoretical Background of Diffusion Models

Diffusion models are generative frameworks that learn to reverse a gradual noising process applied to data. In the forward phase, Gaussian noise is added over many timesteps until the image becomes pure noise, while in the reverse phase, a neural network (typically a U-Net) learns to predict and remove this noise step by step, reconstructing a high-quality image conditioned on a low-resolution input. The forward process gradually adds Gaussian noise:

$$q\left(x_t \middle| x_{(t-1)}\right) = N\left(x_t; \sqrt{(1-\beta_t)}x_{(t-1)}, \beta_t I\right)$$

Here,

- $x_t$is the noisy sample at time t,
- $x_{t-1}$is the sample from the previous timestep,
- β_t is the noise variance added at time t,
- I is the identity covariance matrix.

After T steps, we can directly sample:

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t ; \sqrt{(\bar{\alpha}_t)} \cdot x_0, (1-\bar{\alpha}_t) \cdot I\right)$$

with $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$.

Here,

- $\sqrt{\bar{\alpha}_t} x_0$ represents the remaining clean signal after t noising steps,

- $1 - \bar{\alpha}_t$ defines how much noise dominates the sample at time t.

The reverse process learns to denoise step-by-step using a neural network $\epsilon_\theta(x_t, t)$

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right)$$

Here,

- $\mu_\theta(x_t, t)$ is the predicted mean used to estimate the previous denoised sample,
- $\Sigma_\theta(x_t, t)$ represents the model's learned uncertainty in the reverse step.

The training objective minimizes the prediction error of the noise:

$$L_{simple} = E_{x_0,\varepsilon,t}\left[\left||\varepsilon - \varepsilon_\theta(x_t, t)|\right|^2\right]$$

Here,

- ε is the true noise injected during the forward process,
- $\varepsilon_\theta(x_t, t)$ is the noise predicted by the neural network.

The noise schedule β_t can be linear, cosine, or exponential, controlling the rate of degradation.

The choice of noise schedule critically affects performance: linear schedules add noise at a constant rate, while cosine schedules distribute it more adaptively, gently at the start and end, and more strongly in the middle, yielding clearer intermediate states and improved stability and image quality.

Conditioning further guides the model using auxiliary information such as text, class labels, or low-quality fMRI images, typically integrated through concatenation or feature modulation. Together, these mechanisms make diffusion models highly stable and powerful for super-resolution or synthetic image generation and other tasks requiring detailed, structurally consistent reconstructions.

Beyond the standard denoising diffusion probabilistic model (DDPM) [3], several important variants refine either the sampling process or stability. Denoising diffusion implicit models (DDIM) [4] introduces a deterministic non-Markovian sampling process, removing stochasticity while preserving the same training objective. It reformulates the reverse process as

$$x_{t-1} = \sqrt{\overline{\alpha}_{t-1}}\, x_0 + \sqrt{1 - \overline{\alpha}_{t-1}}\, \varepsilon_\theta(x_t, t)\,,$$

Here,

- x_{t-1} is the reconstructed sample at the previous timestep,
- $\sqrt{\bar{\alpha}_{t-1}}x_0$ represents the retained clean signal,

- $\sqrt{1-\bar{\alpha}_{t-1}}\varepsilon_\theta(x_t,t)$ is the contribution of the predicted noise.

Eliminating random noise injection and thus reducing the required sampling steps (e.g., from 1000 to 50–100) without loss in perceptual quality.

Elucidated diffusion models (EDM) [5] further stabilizes training by redefining the noise parameterization using a continuous signal-to-noise ratio (SNR) schedule, improving convergence and dynamic range.

Meanwhile, score-based generative models (SGMs) [6] view diffusion as solving a stochastic differential equation (SDE) guided by the score function ∇_x log p(x_t), learned via denoising score matching. Their reverse-time SDE formulation,

$$dx = [f(x,t) - g(t)^2 \nabla_x \log p_t(x)]\, dt + g(t)\, d\bar{w} \,,$$

here,

- $f(x,t)$: the drift term governing deterministic dynamics.
- $g(t)$: the diffusion coefficient controlling noise magnitude.
- $\nabla_x \log\, p_t(x)$: the score function (gradient of the log-density).
- $p_t(x)$: the data distribution at time t.
- $d\bar{w}$: the standard Wiener process increment.

Unifies diffusion and Langevin dynamics under a continuous-time framework, allowing adaptive noise control and theoretically grounded stability. Together, these extensions improve efficiency, controllability, and robustness of modern diffusion-based generation.

In this research, the DDPM formulation was chosen due to its conceptual simplicity, proven stability, and widespread adoption in the literature, making it a reliable foundation for 3D medical super-resolution tasks. However, future work could explore more advanced variants such as DDIM, EDM, or score-based continuous formulations, which may further improve sampling efficiency, convergence stability, and perceptual fidelity in high-dimensional medical imaging contexts.

2.2.2 Latent Diffusion Models (LDM)

Latent diffusion models (LDMs) combine an autoencoder with a diffusion model to perform the slow, iterative denoising process in a compact latent space rather than the full image space. The autoencoder compresses the image into a latent representation, diffusion operates efficiently within this space, and the decoder reconstructs the final high-resolution output. This approach dramatically reduces computation while maintaining high generative fidelity, making LDMs well-suited for super-resolution tasks [7].

In practice, latent diffusion redefines the forward noising process not in the pixel domain x_0 but in the encoded latent space $z_0 = E(x_0)$, where E denotes the encoder of a pretrained autoencoder. The diffusion process then evolves as

$$q(z_t \mid z_{t-1}) = \mathcal{N}\big(z_t; \sqrt{1-\beta_t}\, z_{t-1}, \beta_t I\big),$$

here,

- z_{t-1}: latent representation at timestep $t-1$,
- z_t: noisy latent obtained by adding Gaussian noise,
- β_t: noise variance added at timestep t,
- I: identity covariance.

Analogous to the pixel-space case, but operating on lower-dimensional latent tensors. After T steps,

$$q(z_t \mid z_0) = \mathcal{N}\big(z_t; \sqrt{\overline{\alpha_t}}\, z_0, (1-\overline{\alpha_t})I\big),$$

where,

- $\overline{\alpha_t} = \prod_{s=1}^{t}(1-\beta_s)$.
- $\sqrt{\bar{\alpha}_t} z_0$: remaining clean latent signal after t noising steps,
- $1-\bar{\alpha}_t$: cumulative noise contribution after t steps.

The model learns to reverse this process by predicting the injected noise $\varepsilon_\theta(z_t, t)$, minimizing a latent-space variant of the DDPM objective:

$$\mathcal{L}_{LDM} = E_{z_0,\varepsilon,t}[|\varepsilon - \varepsilon_\theta(z_t, t)|^2]$$

here,

- $\varepsilon \sim \mathcal{N}(0, I)$ is the true noise,
- $\varepsilon_\theta(z_t, t)$ is the model's predicted noise,

Since the diffusion occurs in a scaled latent space, the effective data variance is reduced by a factor proportional to the encoder's normalization constant $\sigma_\mathcal{E}^2$, yielding a rescaled noise schedule $\widetilde{\beta_t} = \frac{\beta_t}{\sigma_\mathcal{E}^2}$. This rescaling preserves the same signal-to-noise ratio as in image space while allowing significantly faster sampling and lower memory usage. The decoder D subsequently reconstructs the final image as $\widehat{x_0} = \mathcal{D}(z_0)$ maintaining high perceptual fidelity even under strong compression.

2.3 Image Super-Resolution

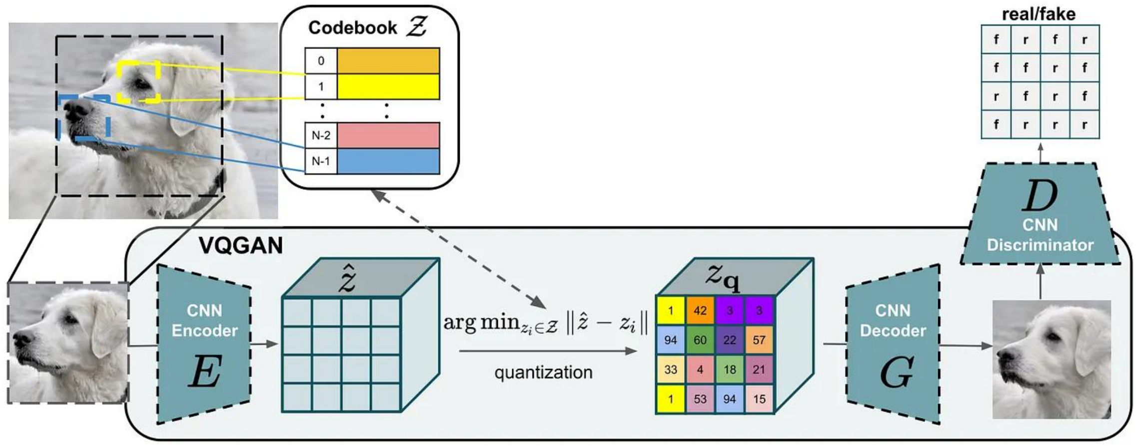
While the clinical utility of super-resolution (specifically for enhancing legacy data and reducing acquisition constraints) was established in the introduction, the technical objective is to solve an ill-posed inverse problem: recovering high-frequency spatial details from low-resolution observations.

In the context of medical imaging, the standard baseline for this task is bicubic interpolation. This traditional upscaling method estimates new voxel values by fitting cubic polynomials across the 16 nearest neighbors of a given point. Although bicubic interpolation is computationally efficient and provides a continuous surface, it inherently acts as a low-pass filter, resulting in overly smooth reconstructions that lack the high-frequency texture required for precise anatomical analysis. Consequently, while inadequate for diagnostic enhancement, bicubic interpolation serves as the quantitative baseline for evaluating the performance of the generative models proposed in this thesis.

2.4 Autoencoders, VAEs, and VQ-GANs

This section will explain the differences between certain autoencoder architectures based on my experiments and research papers.

A critical finding from my initial experiments was the "latent distribution mismatch," where a standard autoencoder produces a continuous and unstructured (deviated) latent space that a generative model struggles to learn. To solve this, my autoencoder evolved into a vector-quantized generative adversarial network (VQ-GAN) [8]. The key difference is the introduction of a vector quantizer layer at the bottleneck. This layer contains a learnable "codebook" of feature vectors. After the encoder produces a latent vector, the quantizer finds the closest vector in its codebook and passes that discrete vector to the decoder

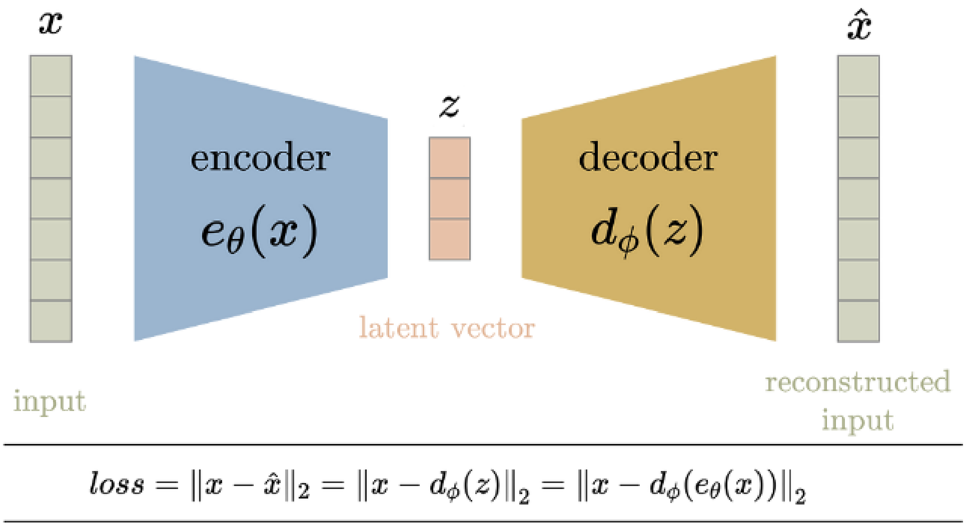


2.1. Figure: VQGAN and the codebook [8]

This has two profound benefits. First, it regularizes the latent space, forcing it onto a structured, discrete manifold that is much easier for the subsequent diffusion model to learn. Second, by combining this with an adversarial training objective, where a discriminator network is trained to distinguish between real and reconstructed patches, we compel the decoder to produce much sharper and more perceptually convincing results than what can be achieved with pixel-wise losses (for instance mean squared error (MSE) or L1) alone.

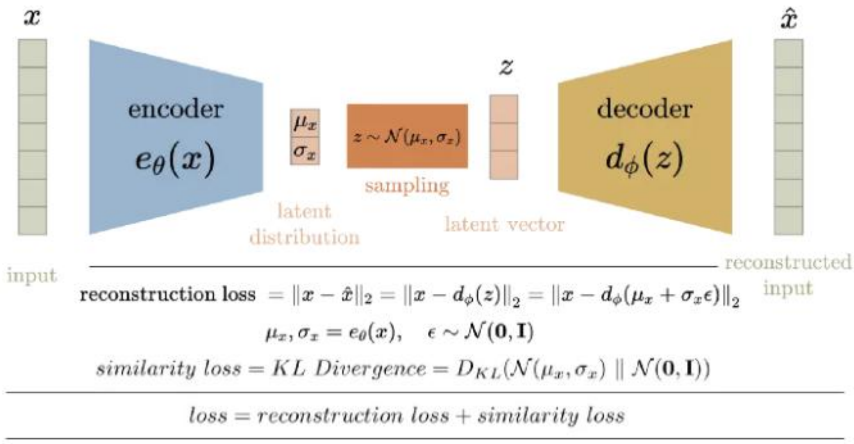
This section delineates the differences between AE, VAE, and VQ-GAN architectures, justifying the selection of VQ-GAN.

Autoencoders (AEs) are powerful tools for compressing images, but they often produce unstable latent spaces, making them unsuitable for generating new, high-quality data.

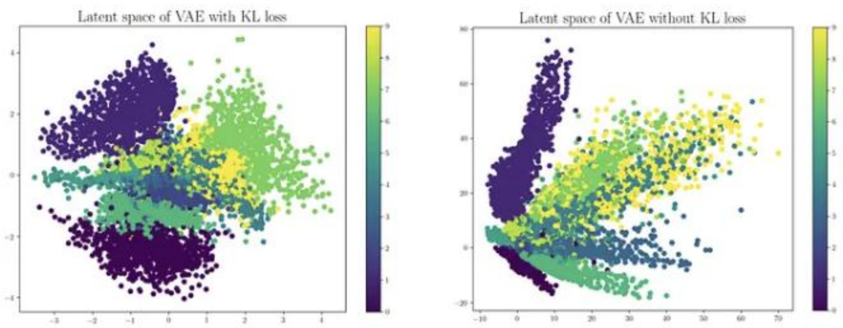


2.2. Figure: Simple AE (autoencoder) [9]

Variational autoencoders (VAEs) improve this by introducing a probabilistic latent space, each input is encoded into a mean and standard deviation, and a latent vector is sampled from this distribution. The KL divergence loss ensures that these latent vectors follow a standard gaussian distribution, helping the model learn a smooth, continuous latent space. However, this often comes at the cost of blurry image reconstructions.



2.3. Figure: Variational autoencoders (VAEs) [9]



2.4. Figure: Difference between the latent spaces' distribution [9]

To address this, vector quantized autoencoders (VQ-VAEs) replace the continuous latent space with a discrete codebook, forcing the model to map inputs to a set of predefined, learnable vectors. VQ-GANs further enhance this idea by adding a GAN-based adversarial loss and a perceptual loss (instead of pixel-wise loss), enabling them to reconstruct images with much sharper details and more realistic textures. For our task (3D fMRI super-resolution) VQ-GAN offers a robust and structured latent space that is both compact and expressive, ideal for downstream generation via diffusion models. It overcomes the blurriness of VAEs and the irregularity of traditional

AEs, making it a significantly better fit for generating realistic high-resolution medical images.

2.4.1 Quantization Error and Codebook Perplexity

An essential aspect of VQ-based architectures is the quality and utilization of the codebook, which can be quantitatively assessed using two key metrics: quantization error and codebook perplexity.

The quantization error measures how closely the encoder's continuous latent outputs $z_e(x)$ align with their nearest codebook entries e_k. Formally, it can be expressed as:

$$\mathcal{L}_{\text{quant}} = ||\text{sg}[z_e(x)] - e_k||_2^2 + \beta||z_e(x) - \text{sg}[e_k]||_2^2$$

where sg[·] denotes the stop-gradient operator and β controls the commitment strength.

Here,

- $z_e(x)$: encoder output before quantization,
- e_k: selected codebook vector (nearest neighbor),
- sg[·]: operator preventing gradients from flowing through its argument,

The first term aligns the codebook with the encoder output, the second term encourages the encoder to "commit" to using its assigned codebook vector. Lower quantization error indicates that the codebook vectors represent the encoded space more accurately, improving reconstruction fidelity and stabilizing downstream diffusion training.

The codebook perplexity, on the other hand, measures how effectively the model utilizes the available codebook entries during training. It is computed as:

$$\text{Perplexity} = \exp\left(-\sum_k p_k \log p_k\right)$$

where p_k is the empirical probability of each codebook vector being selected.

Here:

- p_k: relative frequency of codebook entry $e_k$being chosen,

The expression computes the exponential of the entropy of this distribution,

- high perplexity → many codebook vectors are used uniformly,
- low perplexity → only a few vectors are frequently used (codebook collapse).

A higher perplexity implies more uniform use of the codebook, which correlates with richer latent diversity and reduced overfitting. Conversely, low perplexity (e.g., < 10% of the codebook used) suggests codebook collapse, where only a small subset of embeddings contributes to reconstructions, limiting generative capacity.

In practice, balancing these two metrics is crucial: minimizing quantization error improves reconstruction accuracy, while maintaining high perplexity ensures representational richness in the latent space, both of which directly influence the fidelity and generalization of the diffusion model operating on these latents [10] [8].

3 Materials and Methods

This section outlines the experimental framework, detailing the dataset, preprocessing pipelines, and computational strategies employed for fMRI synthesis and super-resolution.

3.1 Dataset and Preprocessing

3.1.1 Dataset Acquisition

The study utilizes the WU-Minn Human Connectome Project (HCP) 1200 Subjects dataset, a standard reference for high-resolution functional Magnetic Resonance Imaging (fMRI). The data, accessible via the HCP portal, consists of 4D NIfTI volumes recording blood-oxygen-level-dependent (BOLD) signals during five distinct motor tasks: left/right hand, left/right foot, and tongue movements. [11]

3.1.2 Temporal Reduction

Raw fMRI data is inherently four-dimensional (space + time). To adapt this for 3D generative modeling, the time dimension was averaged across the 16-second activation windows for each task. This creates a single, high-quality 3D structural volume per task, preserving spatial activation patterns while reducing computational complexity and noise.

It is acknowledged that averaging the time dimension removes the temporal dynamics required for functional connectivity analysis. However, this step is necessary to reduce the computational complexity for this 3D spatial super-resolution study, effectively treating the input as a static T2*-weighted structural volume.

3.1.3 Patch-Based Preprocessing

Processing full 3D brain volumes exceeds the memory capacity of standard GPUs. Therefore, an offline patch-based strategy was implemented [12] [13]:

- **Normalization**: Volumes were normalized to an intensity range of [0, 1].
- **Patch Extraction**: Using the MONAI library, volumes were cropped into 64*64*64 voxel patches. The *RandCropByPosNegLabeld* transform was

employed to ensure patches were centered on brain tissue rather than background air.

- **Storage**: Patches were serialized to disk to decouple preprocessing from training, eliminating CPU bottlenecks.

3.2 Model Training Strategies

3.2.1 Skip Connections and Student-Teacher Learning

A fundamental challenge in latent diffusion models (LDMs) is the trade-off between reconstruction fidelity and generative capability. Autoencoders typically rely on skip connections to transfer high-frequency spatial details ("where" information) from the encoder to the decoder [14] [15]. However, during generative inference, the diffusion model produces a latent vector without a corresponding encoder input, meaning skip connections are unavailable [7] [8].

To resolve this, a Student-Teacher training paradigm was adopted:

- **Teacher Network**: A standard autoencoder with skip connections is trained to produce high-fidelity reconstructions.
- **Student Network**: A parallel "no-skip" decoder learns to mimic the teacher's output using only the latent bottleneck. This effectively distills the spatial information into the latent space, allowing the student model to generate sharp images during inference without relying on skips [16] [17].

3.3 Classifier-Free Guidance (CFG)

To improve adherence to the low-resolution conditioning inputs, classifier-free guidance (CFG) was implemented during diffusion training. The model is jointly trained on conditional and unconditional objectives by randomly dropping the condition during training. During inference, the prediction is a weighted combination of both:

$$\hat{\epsilon} = \epsilon_\theta(x_t|\emptyset) + s \cdot \left(\epsilon_\theta(x_t|c) - \epsilon_\theta(x_t|\emptyset)\right)$$

where s is the guidance scale. This amplifies the signal of the conditioning image c without requiring a separate classifier model. [18]

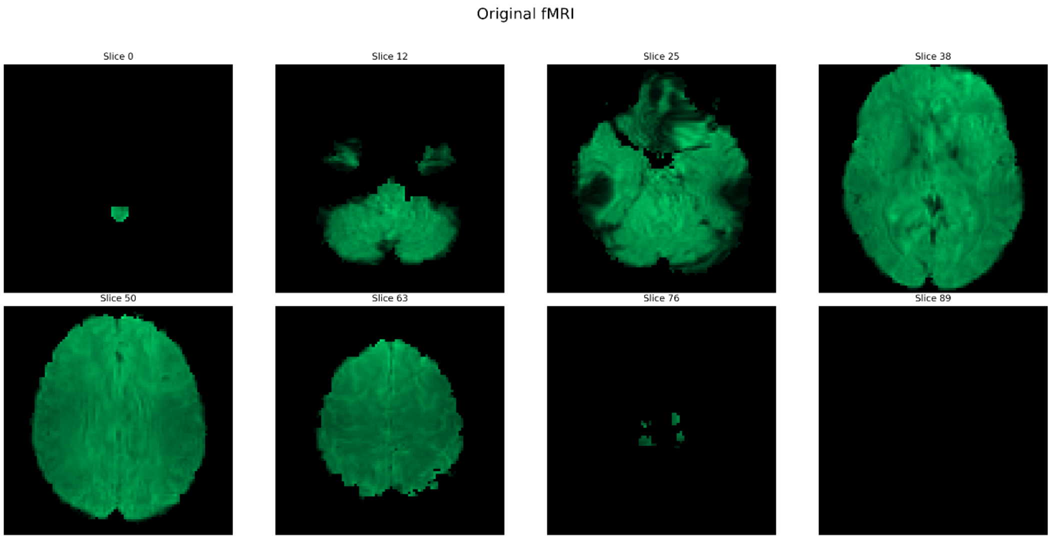
3.4 Adversarial Fine-Tuning

To further enhance perceptual sharpness, the decoder was fine-tuned using an adversarial objective. A discriminator network was introduced to distinguish between real fMRI patches and reconstructions [19]. This "minimax" game forces the generator to produce texture-rich outputs that produces better statistical realism, overcoming the blurring effect often associated with pure mean squared error (MSE) losses. [7] [20]

3.5 Inference and Evaluation

3.5.1 Sliding Window Inference

Full-volume reconstruction is performed using sliding window inference, since patched-based learning was used for memory efficiency. The model processes the volume in overlapping patches, which are then stitched together. Gaussian weighting is applied to the overlap regions to blend patch boundaries seamlessly, ensuring global structural consistency. To visualize this Figure 3.1 represents an fMRI image with sliding window inference.



3.1. Figure: fMRI image, with sliding window inference

3.6 Evaluation Metrics

Model performance is evaluated against a Bicubic Interpolation baseline using standard image quality metrics.

3.6.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR quantifies the reconstruction quality by measuring the ratio between the maximum signal power and the noise introduced during generation. Expressed in decibels (dB), higher values indicate a closer approximation to the ground truth. In the context of fMRI, PSNR is a critical measure of pixel-wise fidelity, ensuring that the super-resolved output maintains statistical consistency with the original scan. While a traditional metric, it remains a robust proxy for measuring fidelity in low-level vision tasks [21].

3.6.2 Structural Similarity Index Measure (SSIM)

SSIM evaluates the perceptual quality of the image by calculating the similarity in luminance, contrast, and structure between the reconstruction and the reference. Unlike PSNR, which relies on absolute pixel differences, SSIM focuses on the inter-dependencies of spatially adjacent pixels. This is particularly relevant for medical imaging, as it better captures the preservation of anatomical boundaries and functional structures [22]. The index ranges from -1 to 1, where a value of 1 indicates identical structural integrity.

3.6.3 Limitations of Traditional Metrics

It is acknowledged that traditional metrics like PSNR and SSIM have limitations in aligning with human perception. Specifically, it has been mathematically proven that minimizing distortion (MSE/PSNR) inherently leads to a degradation in perceptual quality (sharpness/texture), a phenomenon known as the perception-distortion tradeoff [23]. This often results in metrics penalizing realistic, texture-rich outputs while favoring smoother, blurrier reconstructions.

While learned perceptual metrics such as LPIPS [24] or DISTS [25] offer better alignment with human vision by comparing deep features. they can be sensitive to preprocessing artifacts and often require interpretation that exceeds the diagnostic focus of this study [26]. Therefore, this thesis relies on PSNR and SSIM to provide a standardized, reproducible benchmark for structural and signal fidelity.

3.7 Computational Resources

All models were implemented in PyTorch and MONAI, utilizing Hugging Face Diffusers for noise scheduling. Training was conducted on a high-performance cluster provided by Artilance Kft., utilizing an NVIDIA RTX 3090 (24 GB VRAM) and 128 GB RAM to handle 3D volumetric data.

4 Implementation and Experiments

4.1 Data Loading and Preprocessing Pipeline

To manage the large-scale fMRI dataset, a structured pipeline was implemented using the PyTorch Lightning framework. This ensures reproducible data handling by standardizing the loading of NIfTI volumes and managing stratified training, validation, and test splits.

A critical optimization was introduced for the super-resolution task. Early experiments revealed that generating patches dynamically during training caused significant CPU bottlenecks. To address this, the pipeline was restructured to utilize offline-preprocessed 64×64×64 patches stored on disk (see Section 3.1.3). This shift from dynamic to static loading eliminated runtime overhead, maximizing GPU throughput and facilitating scalable experimentation.

4.2 Preliminary Experiment: Synthetic fMRI Generation

4.2.1 Autoencoder Architecture

To compress high-dimensional fMRI data into a manageable latent representation, a 3D autoencoder was implemented. The encoder comprises three sequential 3D convolutional layers with channel dimensions scaling from 1 to 128 ($1 \rightarrow 32 \rightarrow 64 \rightarrow 128$). Each layer utilizes a kernel with a stride of 2 and padding of 1 to progressively halve the spatial resolution while preserving boundary features. ReLU activations were applied after each convolution to ensure efficient gradient propagation. This architecture reduces the original input volume of (1, 91, 109, 91) to a feature tensor of (128, 12, 14, 12). This tensor is flattened and passed through a fully connected layer to produce a compact 256-dimensional latent vector. The decoder mirrors this structure using 3D transposed convolutions with output padding to ensure the reconstructed volume aligns perfectly with the input dimensions. To incorporate task-specific information, label embeddings are concatenated directly with the latent vector prior to decoding, preserving the latent structure while enabling conditional generation.

4.2.2 Diffusion Model

The generative component utilizes a standard UNet2DModel from the Hugging Face Diffusers library, selected for its optimization and stability. To adapt the 3D data for this 2D architecture, the 256-dimensional latent vector derived from the autoencoder is reshaped into a 16*16 2D pseudo-image. This approach significantly reduces computational overhead compared to native 3D diffusion.

The U-Net employs a hierarchical structure with DownBlock2D and UpBlock2D modules for feature extraction and reconstruction. To capture global dependencies within the latent representation, AttnDownBlock2D and AttnUpBlock2D modules are integrated. Group normalization is utilized instead of batch normalization to maintain training stability given the small batch sizes required by the hardware constraints. Similar to the autoencoder, conditioning is achieved by concatenating learnable label embeddings with the latent input.

4.2.3 Training and Optimization

The autoencoder and diffusion model were trained sequentially to decouple the representation learning from the generative process and to adhere to GPU memory limitations.

- **Autoencoder:** The model was trained for 50 epochs (patience: 10) using a composite loss function to balance pixel-wise accuracy and structural fidelity: $\mathcal{L} = (\text{MSE}) + 0.1 \cdot \big(1 - \text{structural similarity index measure (SSIM)}\big)$. Initial experiments using only MSE resulted in insufficient structural detail.
- **Diffusion Model:** The model was optimized for 100 epochs using a standard Mean Squared Error (MSE) loss to predict noise residuals. A Denoising Diffusion Probabilistic Model (DDPM) scheduler was employed with 1,000 timesteps to regulate the forward and reverse diffusion processes.

Subsequent experiments suggested that incorporating L1 loss and increasing the weight of the SSIM component in the autoencoder training significantly improves structural sharpness in the final output.

4.3 Experiment 2: fMRI Super-Resolution

Building upon the insights from the initial generative experiments, it became evident that generating high-fidelity medical images from a purely random latent space presents significant challenges, particularly in preserving fine-grained anatomical detail. This section pivots from unconditional generation to the more constrained and clinically impactful task of super-resolution. This approach leverages the same foundational technologies (a powerful autoencoder and a latent diffusion model) but applies them to enhance existing low-resolution data rather than creating new data from scratch. The following sections will detail the development of a state-of-the-art super-resolution pipeline.

4.3.1 Autoencoder

The first and most critical stage of my latent diffusion model (LDM) is the development of a powerful autoencoder. Its primary purpose is not merely to reconstruct images, but to learn a compressed, semantically rich, and structurally organized latent space. This latent space forms the foundation upon which the subsequent diffusion model will operate. My design choices were heavily informed by initial experiments, which revealed the shortcomings of simpler architectures, and are aligned with state-of-the-art practices in generative medical imaging.

4.3.1.1 Autoencoder Architecture and Rationale

My final autoencoder architecture is a fully convolutional 3D U-Net, similarly to previous solutions, a design pattern renowned for its efficacy in medical imaging tasks that require preserving fine spatial details [15].

The encoder path progressively down-samples the input volume through a series of DownBlock3D modules, each containing ResidualBlock3Ds, and a stride = 2 convolution to halve the spatial dimensions while increasing the number of feature channels. The core neural network modules such as convolutional and residual blocks were custom-implemented in PyTorch for maximum control. Symmetrically, the decoder path uses UpBlock3D modules with transposed convolutions to up-sample the feature maps back to their original resolution using a scale factor of 2 and trilinear interpolation.

The core of this architecture is built using modern components:

- **GroupNorm** (group normalization) is used for normalization to ensure stability with small batch sizes
- **SiLU** (sigmoid linear unit) is used as the activation function for its smooth, non-monotonic properties that benefit gradient flow.

To make this implementation possible, PyTorch was leveraged for its flexibility and MONAI, an open-source framework for deep learning in healthcare imaging. MONAI was indispensable for handling 3D data, particularly for implementing the patch-based training strategy, The MONAI framework was utilized for two critical and computationally complex aspects of the pipeline: data handling and inference.

4.3.1.2 Network Blocks and Channel Configuration

The foundational building blocks of my U-Net, namely the ResidualBlock3D and the DownBlock3D/UpBlock3D modules, were custom-implemented. While pre-built blocks are available in libraries, a custom implementation provided the necessary flexibility to integrate my switchable skip-connection logic and specific normalization choices.

A base channel count of 32 was chosen. The encoder progressively doubles this channel count at each of its three down-sampling stages, following a standard convention: 32 → 64 → 128 → 256. This design allows the network to capture increasingly complex and abstract features as the spatial resolution decreases.

The number of channels is a critical hyperparameter that balances model capacity and VRAM usage. A base of 32 provided a sufficient capacity for learning fMRI features while remaining trainable on my 24GB GPU. The final bottleneck layer projects the deep 256-channel feature map into a compact 8-channel latent representation. This results in a final discretized latent volume of $8 \times 8 \times 8 \times 8$, achieving the significant compression ratio required for the LDM.

4.3.1.3 Vector Quantizer Implementation

This section examines in detail how the vector quantizer solution was implemented. The vector-quantizer layer is a custom implementation based on the original VQ-VAE paper [10].

At its core is a learnable "codebook," which is essentially a PyTorch nn.Embedding layer containing N embedding vectors (in this implementation, $N = 512$), each of dimension $D = 8$. When the encoder produces a continuous latent tensor z, the quantizer calculates the L2 Euclidean distance between each spatial feature vector in z and all N codebook vectors. It then replaces each feature vector in z with its nearest neighbor from the codebook, producing the quantized latent z_q.

This process maps the continuous latent space to a discrete set of learned prototypes. The model is trained to minimize a composite objective function consisting of two parts (see in detail in section 2.4.1):

- **Codebook Loss**: $||\mathrm{sg}[z] - z_q||_2^2$, which pulls the codebook vectors toward the encoder outputs.
- **Commitment Loss**: $\beta||z - \mathrm{sg}[z_q]||_2^2$, which encourages the encoder's output to stay "committed" to the chosen codebook vectors.

Formally, the total vector-quantization loss is the sum of the 2 expressions. Minimizing the codebook loss updates the dictionary to represent the data better, while minimizing the commitment loss prevents the encoder's output from fluctuating too wildly between codebook vectors, thus stabilizing the latent space for the downstream diffusion model.

4.3.1.4 Patch-Based Training for Memory Efficiency

Training deep 3D convolutional networks on full-sized fMRI volumes (e.g., 91x109x91) are computationally infeasible on consumer-grade GPUs due to enormous VRAM requirements.

To overcome this, a patch-based training approach was adopted, which is significantly different compared what was used before for section 4.2. As explained in more depth in section 3.1.3, instead of feeding the entire volume to the model, my data-loader extracts smaller, random 3D chunks or "patches" (e.g., 64x64x64) from the original images. To only extract relevant parts of the images a pre-implemented solution was used from the MONAI library for this task.

The model is trained exclusively on these patches. This is a highly effective and standard practice because convolutional networks learn translation-invariant features (features that can be recognized even if they move around slightly in the image) by seeing thousands of diverse local patches from all over the brain, the model learns the fundamental patterns, textures, and structures of fMRI data.

During inference on a full volume, a sliding window technique was used (see in section 3.5.1), processing the image patch by patch and smoothly blending the overlapping results to form a coherent, full-sized reconstruction, for this task MONAI's solution was leveraged again. This allows us to train a powerful, deep model on high-resolution data while staying within my hardware constraints [27].

4.3.2 Diffusion Model

The second stage of my Latent Diffusion Model (LDM) pipeline is the diffusion model itself, which forms the generative core of the system. Unlike the autoencoder, which learns a deterministic mapping from image to latent and back, the diffusion model learns a stochastic, iterative process to generate data from pure noise (see Section 2.2, Theoretical Background of Diffusion Models). My implementation is a conditional 3D U-Net, specifically designed for super-resolution, which leverages the latent representation of a low-resolution input as a guide to reconstruct a high-resolution output.

4.3.2.1 Diffusion Model Architecture

The architecture follows the structure of a 3D U-Net, composed of residual blocks [28] and skip connections, but adapted for iterative denoising. Two key elements enable conditional super-resolution:

- **Conditional Input:**
  The model is designed for a conditional task. Its input is not just the noisy latent it needs to denoise (z_{hr_noisy}), but also the conditioning information from the low-resolution image ($z_{lr\_upsampled}$). These are concatenated along the channel dimension, effectively doubling the number of input channels. This allows the U-Net to learn to use the features from the z_{lr} channels as a direct spatial guide for denoising the z_{hr} channels.

- **Time Embeddings:**
  Diffusion models must be aware of the current noise level. The timestep t is converted into a high-dimensional sinusoidal embedding, similar to positional encodings used in Transformers. [29] This embedding is processed through a

small multilayer perceptron (MLP) and injected into each residual block, enabling the model to adapt its denoising behavior to different levels of noise.

To ensure precise control over the generative process, noise scheduling and sampling are managed through a custom squared cosine schedule and a manual DDPM sampling loop. This implementation ensures stable diffusion dynamics tailored specifically to the volumetric distribution of the fMRI latent space without reliance on external inference libraries.

4.3.2.2 Training Process

The training objective is to teach the diffusion model to predict the noise that was added to a high-resolution latent. This ensures that the network learns the reverse diffusion process while respecting the structure of the low-resolution guide.

For each high-resolution patch:

- **Create Paired Data:**
  A corresponding low-resolution patch (p_{lr}) is created on-the-fly by synthetically degrading the p_{hr} with 3D trilinear down-sampling.

- **Encode to Latent Space:**
  Both patches are passed through the frozen, pre-trained autoencoder to obtain z_{hr} (the high-quality target latent) and z_{lr} (the low-quality guide latent).

- **Simulate Forward Diffusion:**
  A random amount of Gaussian noise ε is added to z_{hr} to produce z_{hr_noisy}. This corresponds to the forward diffusion process described theoretically in Section 2.2.1.

- **Calculate Loss:**
  The conditional U-Net predicts the noise, and training minimizes the MSE between the predicted and actual noise. This follows the DDPM principle that diffusion models do not learn to reconstruct images directly, but to estimate the noise added during the forward diffusion process.

4.3.3 Finetuning

The fine-tuning of the pipeline followed a multi-stage progression designed to align the diffusion model's latent generations with the decoder's reconstruction capabilities. The baseline configuration consisted of a vector quantized variational autoencoder (VQ-VAE) [10], trained to establish a discrete latent space, alongside a conditional diffusion model (DDPM) trained to predict noise within that space [3]. This framework was selected as it has proven highly effective for super-resolution tasks [30]. To achieve perceptual realism and structural fidelity, four specific strategies were implemented to address the "missing skip connection" problem inherent in latent diffusion models.

4.3.3.1 Dual-Path Decoder Architecture

A fundamental challenge in this architecture is the availability of skip connections. Autoencoders, particularly in biomedical imaging, typically rely on skip connections to transfer high-frequency spatial details ("where" information) from the encoder to the decoder [14]. However, during generative inference, the diffusion model produces a latent vector without a corresponding encoder input, meaning skip connections are unavailable [7].

To address this discrepancy, the decoder architecture was extended with a dual-path strategy. Two distinct convolutional paths were implemented within the up-sampling blocks:

- **Path A (Skip-Enabled):** Accepts concatenated skip connections from the encoder.
- **Path B (Skip-Free):** Operates independently without external skips.

This design allows the model to learn an expressive latent space using Path A, while simultaneously being fine-tuned to handle the skip-free condition via Path B. While it is theoretically possible to use skip connections derived from the degraded low-resolution input, studies suggest that relying on features from lossy inputs can propagate noise rather than restoring structure [14]. Therefore, the dual-path approach ensures the model relies on the latent representation rather than noisy artifacts.

4.3.3.2 Student–Teacher Distillation

To further robustify the skip-free path, a knowledge distillation strategy was implemented [16]. In this setup, the skip-enabled decoder acts as a fixed "teacher," providing high-quality reference reconstructions. The skip-free decoder acts as a "student," trained to approximate the teacher’s output using only the latent bottleneck.

The training objective combined standard voxel-level losses (L1 and MSE) with perceptual loss and feature-matching terms [17]. This encourages the student to replicate the structural sharpness of the teacher, effectively "distilling" the spatial information usually carried by skip connections into the latent space.

4.3.3.3 Classifier-Free Guidance (CFG) Alignment

To ensure the diffusion model generates latent vectors that fall within the VQ-VAE's valid distribution, classifier-free guidance (CFG) was applied. During training, a 10% conditioning dropout rate was introduced, forcing the model to learn both conditional and unconditional denoising [18]. During inference, a guidance scale of 7.5 was used to interpolate between these predictions. This not only improves adherence to the low-resolution conditioning but also implicitly reduces distribution mismatches between the generated latents and the decoder's expected input.

4.3.3.4 Adversarial Fine-Tuning (VQ-GAN)

The final stage focused on maximizing perceptual realism (texture and sharpness) by adversarially fine-tuning the autoencoder, effectively converting it into a VQ-GAN [8]. A custom 3D discriminator was trained from scratch to distinguish between real fMRI patches and model reconstructions [19].

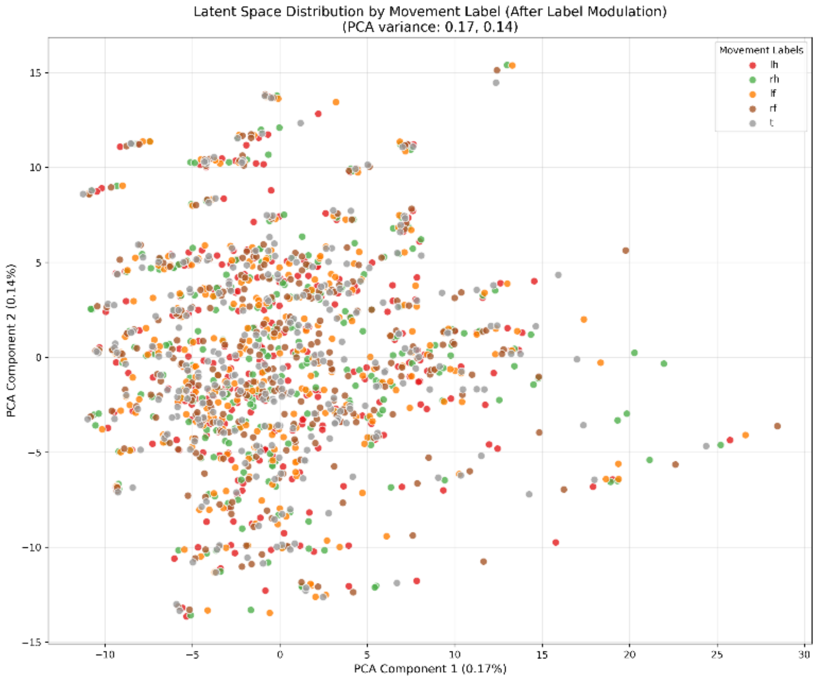
Initial experiments training exclusively on the skip-free path resulted in "catastrophic forgetting," where the model lost its ability to reconstruct valid anatomy [31]. To stabilize this, a composite loss function was introduced, combining the adversarial loss with an anchoring loss to preserve the original reconstruction capabilities. Following best practices for GAN stability, the learning rates were set to 10^{-5} for the autoencoder and 5×10^{-5} for the discriminator, with a reduced adversarial loss weight of 0.001. This balanced approach yielded sharp, anatomically plausible reconstructions without introducing hallucinated artifacts [32]. Results

4.4 Results for the fMRI Synthesis

Before this section, it is important to note that the thesis' images are high-quality. Any blurriness you see in the outputs is not due to the image quality itself, but rather a limitation of the generative model. In other words, if a result appears blurry, the original image taken was sharp, the model's reconstruction or generation ability was simply not strong enough.

4.4.1 Label Embedding and Its Flaws

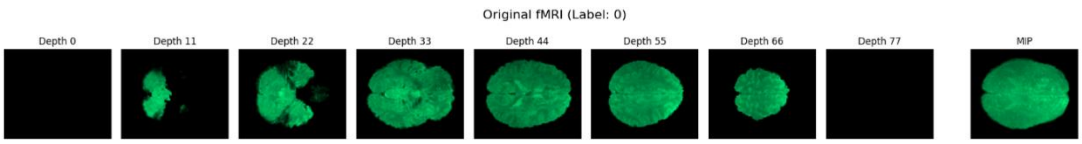
The label embedding strategy failed to induce class-conditional generation. Neither simple concatenation nor adaptive modulation produced distinguishable outputs across labels. Latent space analysis (Figure 4.1) confirms that samples cluster by subject rather than task. This indicates that while structural fMRI features were preserved, the conditioning signal was insufficient, suggesting a need for more expressive multi-layer modulation.



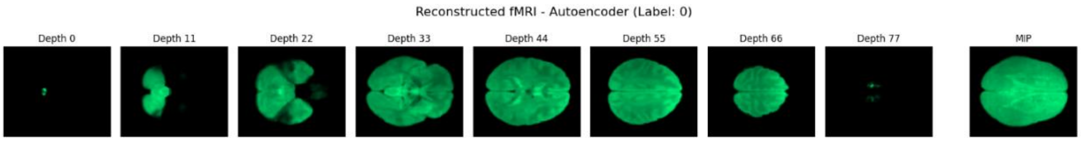
4.1. Figure: Latent space visualization by labels (PCA)

4.4.2 Results For The Synthetic fMRI Autoencoder

Comparison of the original and reconstructed images (Figures 4.2 and 4.3) reveals that while global brain structure is preserved, the output suffers from significant blurring. This suggests the 256-dimensional latent space created a severe information bottleneck.

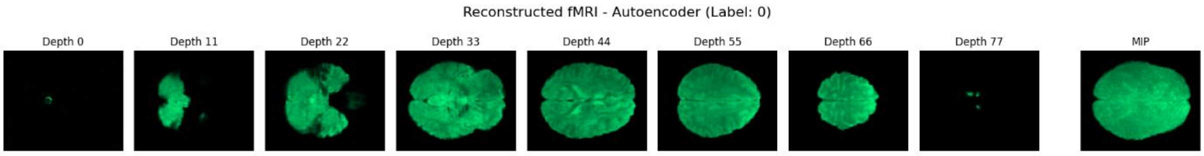


4.2. Figure: Original image



4.3. Figure: Reconstructed image

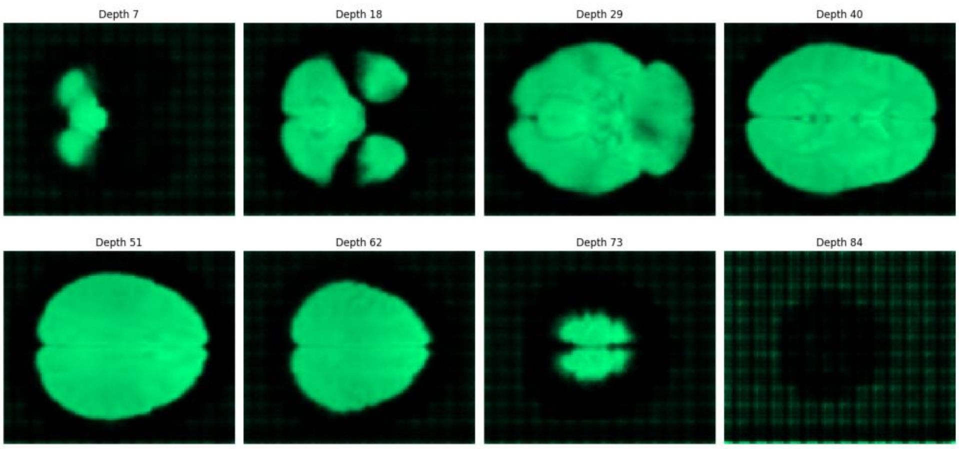
To address this, a revised architecture was implemented featuring a spatially-aware 8×8×8 3D latent space, group normalization, and skip connections. While this approach yielded sharper reconstructions (Figure 4.4), the reliance on skip connections proved incompatible with the diffusion framework. Since generative inference lacks the encoder pass required to populate these connections, the decoder failed to function correctly. Attempted mitigation strategies, including zero-filling and predictive networks, were unsuccessful (Figure 4.9).



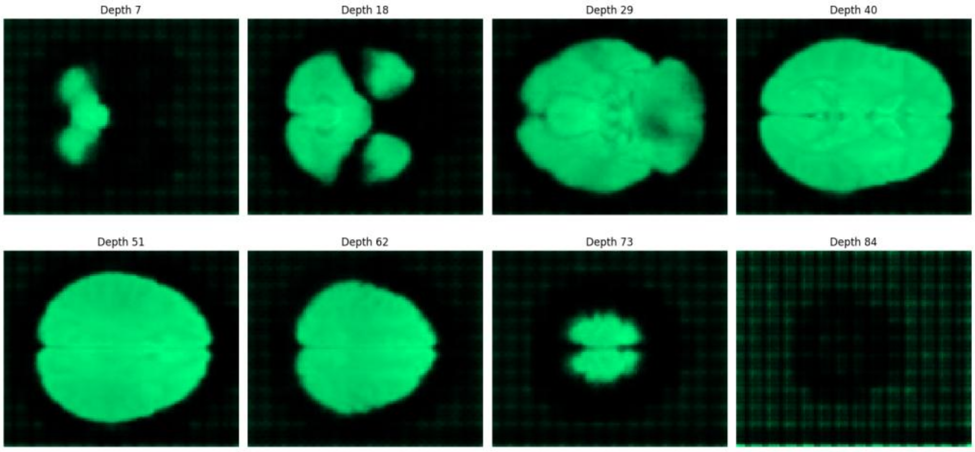
4.4. Figure: Reconstructed image by the newly introduced autoencoder

4.4.3 Results For The Synthetic fMRI Diffusion Model (4.2.2)

While the initial approach offered computational efficiency, it proved insufficient for high-fidelity generation. As evidenced by Figures 4.5 and 4.6, outputs conditioned on different labels ("left hand" vs. "right foot") are virtually indistinguishable, indicating that the label embedding failed to influence generation. Furthermore, the generated images lack high-frequency detail, inheriting the blurring artifacts observed during the autoencoder stage.



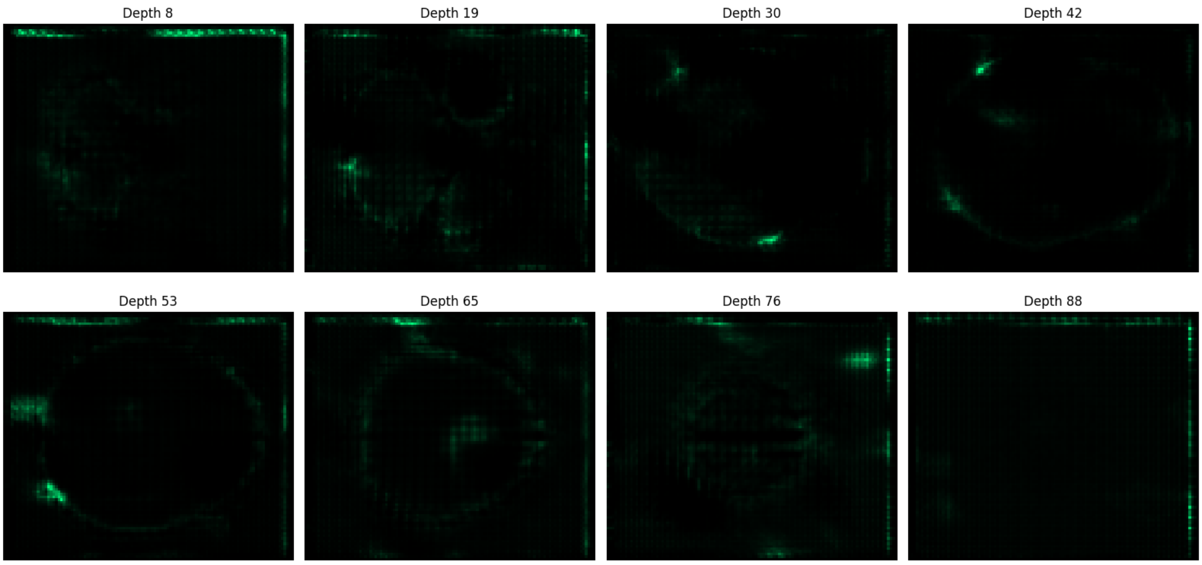
4.5. Figure: Generated image representing a "left hand" movement



4.6. Figure: Generated image representing a "right foot" movement

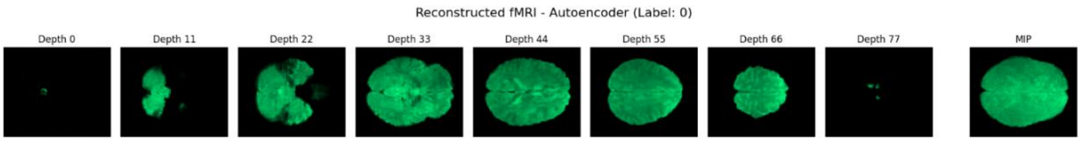
To address these limitations, a custom 3D U-Net architecture was implemented. This model utilized 3D convolutions (1 → 256 channels), additive time-step embeddings, and a cosine noise schedule, which showed empirical improvements over linear scheduling. Additionally, Sigmoid Linear Unit (SiLU) activations were adopted to prevent dead neurons and ensure smoother gradient flow.

Despite these architectural refinements, the generated samples (Figure 4.7) exhibited severe structural degradation. The primary cause was identified as the misalignment between the generated latents and the decoder requirements.

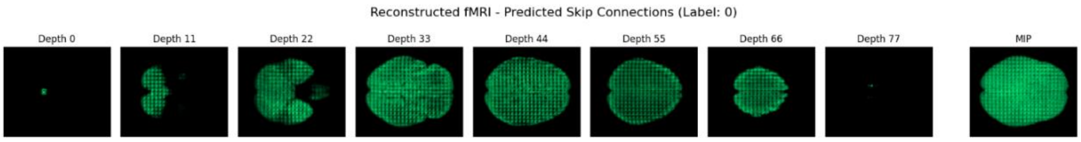


4.7. Figure: Generated "left hand" image with the experimental 3D setup

As shown in Figure 4.9, attempting to predict skip connections from the latent space resulted in significant information loss and artifacts compared to using the original skip-connections (Figure 4.8). Consequently, while the specific 3D architecture was discarded, effective components such as SiLU and cosine scheduling were retained for the super-resolution models discussed in later sections.



4.8. Figure: Reconstructed image with the original skip connections



4.9. Figure: Reconstructed image with predicted skip connections

4.4.4 Quantitative Evaluation and Latent Distribution Analysis

Quantitative metrics reveal a disconnect between pixel-wise accuracy and structural fidelity. While the autoencoder achieved a low Test MSE (0.0054) and L1 loss (0.040), the SSIM score of 0.48 remains low, correlating with the lack of structural sharpness observed in the visual outputs. This confirms that pixel-level minimization alone is insufficient for preserving high-frequency fMRI details.

Critically, analysis of the latent space reveals a significant distribution mismatch. The generated latent vectors exhibit a standard deviation of 0.75, compared to 4.30 in the original encoder distribution. This variance collapse indicates that the diffusion model failed to capture the full expressivity of the latent space. These findings suggest that standard MSE loss is inadequate for latent alignment and motivate the introduction of KL-divergence regularization (VAE) in the subsequent super-resolution architecture.

4.5 Results for the fMRI Super-Resolution

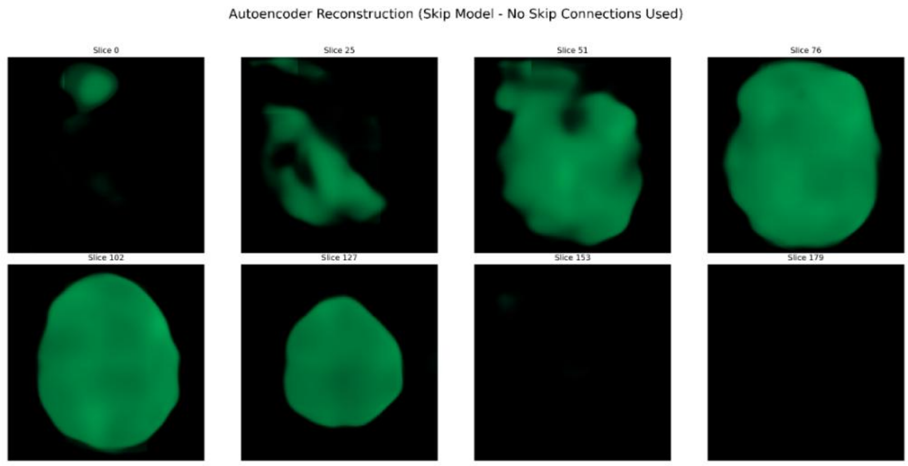
This section evaluates the proposed Latent Diffusion Model (LDM) pipeline for fMRI super-resolution. The experimental trajectory addressed three critical challenges:

- The misalignment of generated latents with the encoder's distribution,
- The decoder's dependency on skip connections,
- The trade-off between perceptual texture and signal fidelity.

Quantitative results are summarized in Table 4.1.

4.5.1 Autoencoder Bottlenecks

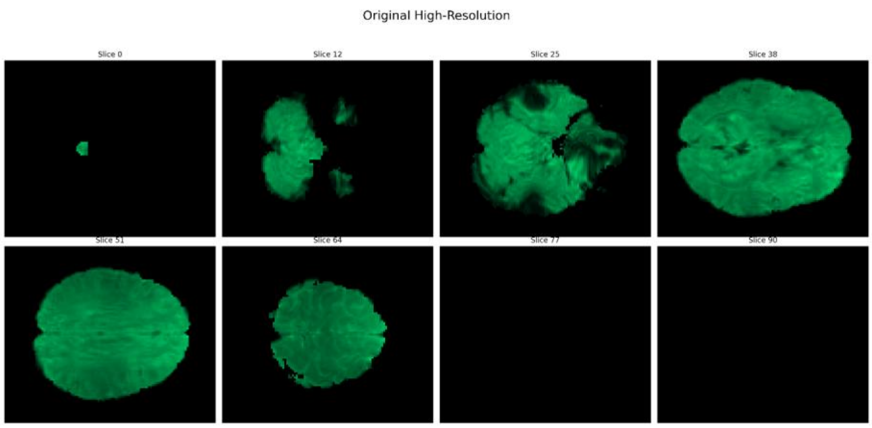
The VQ-VAE achieves near-perfect reconstruction when skip connections are active (PSNR: 67.14 dB). However, generative inference precludes the use of encoder-side skip features. Without these connections, the decoder’s performance collapses to 24.02 dB PSNR (0.66 SSIM), falling below the bicubic interpolation baseline (27.73 dB). This identifies the decoder's lack of spatial context as the primary system bottleneck.



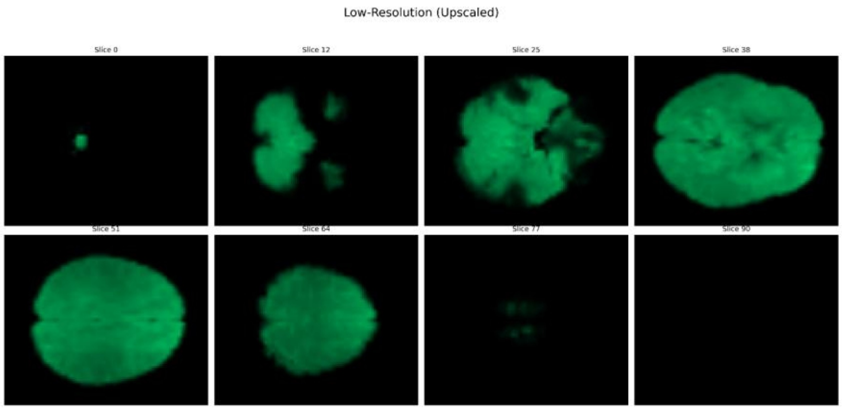
4.10. Figure: Student-Teacher finetuned autoencoder simple reconstruction without skip connections

4.5.2 Diffusion Model and Latent Alignment

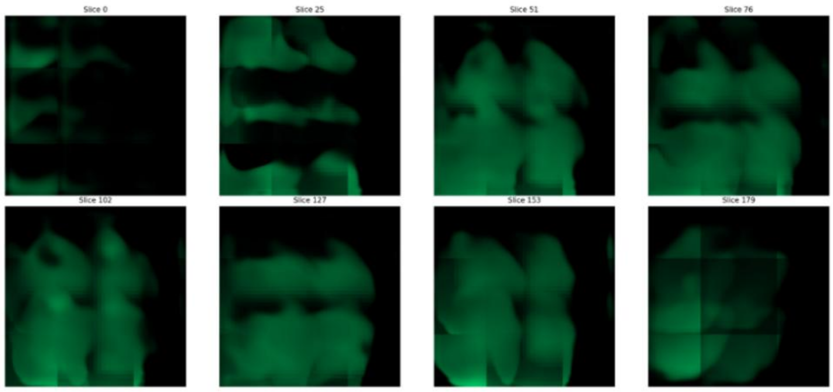
Initial experiments with the LDM resulted in severe structural degradation (18.87 dB PSNR). Analysis of the latent space revealed a latent distribution shift of 27.18%, indicating that the diffusion model generated feature vectors outside the VQ-VAE's valid manifold.



4.11. Figure: Original image

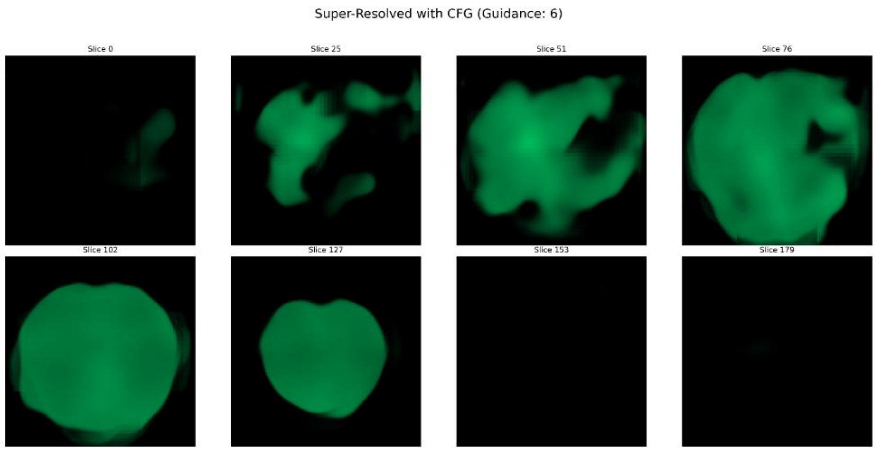


4.12. Figure: Low resolution image as condition



4.13. Figure: Enhanced image (SR) after finetuning (student-teacher) the decoder.

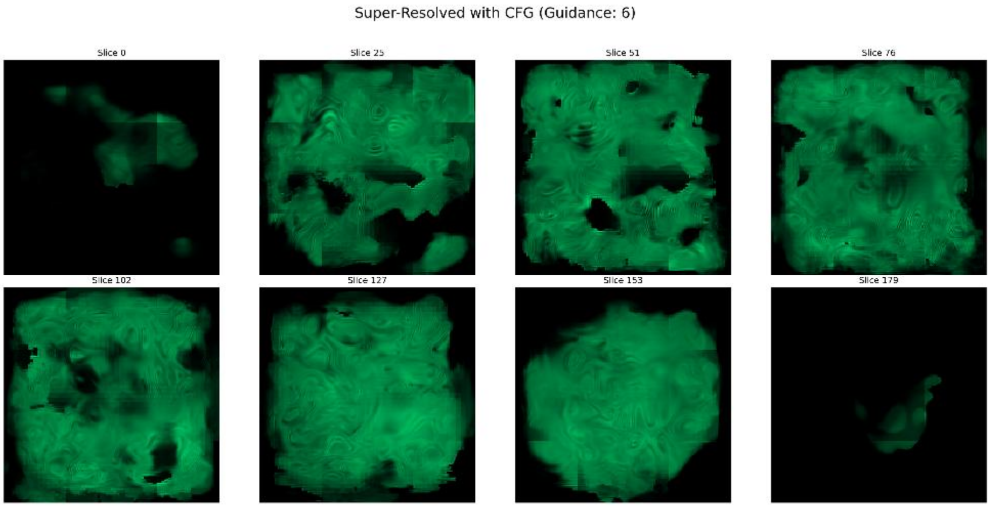
The implementation of Classifier-Free Guidance (CFG) significantly mitigated this issue. Although CFG was primarily adopted to improve conditioning fidelity, the associated retraining indirectly rectified the latent space misalignment, dropping the distribution shift to 0.12% (PSNR: 23.56 dB). While this solved the validity of the generated latents, the resulting images remained overly smooth, limited by the decoder.



4.14. Figure: SR image with the CFG finetuned diffusion model

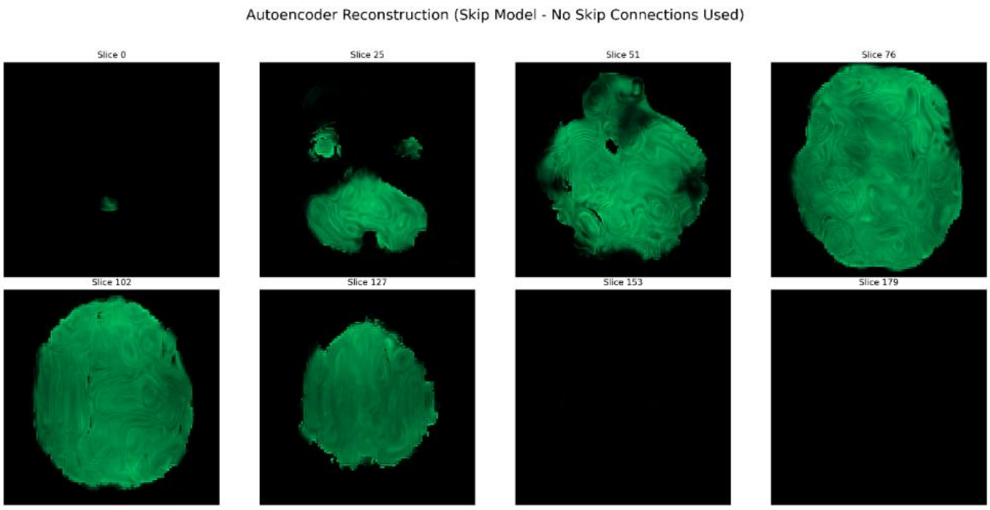
4.5.3 Adversarial Fine-Tuning of the Decoder

To recover high-frequency spatial details, the decoder was fine-tuned using an adversarial framework. The initial unanchored adversarial training led to catastrophic forgetting, degrading PSNR to 11.17 dB (see figure 4.15).

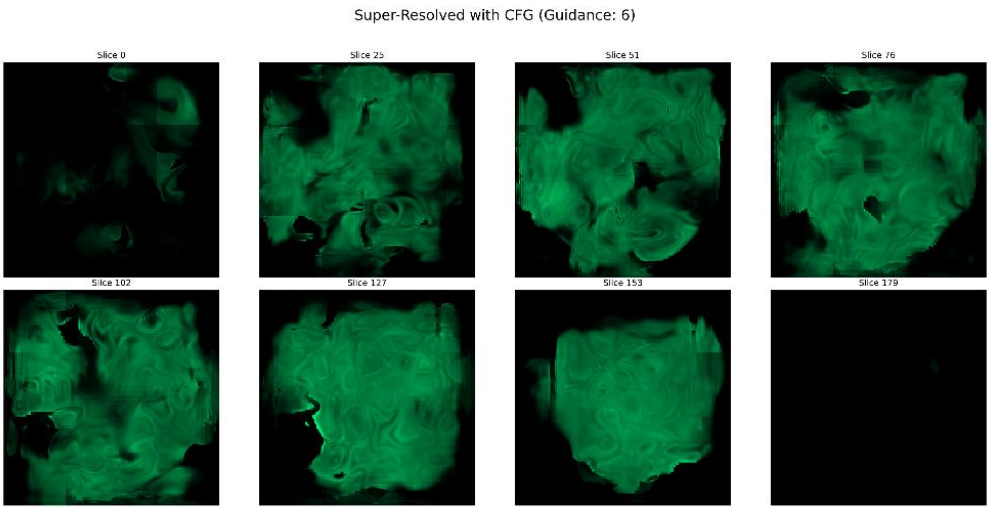


4.15. Figure: SR with the first adversarial finetuning attempt

Introducing an L1 anchoring loss stabilized training (Table 4.1, row 5), yielding sharper textures, especially for pure reconstruction (Figure 4.16). However, this introduced characteristic GAN artifacts (e.g., unnatural curvature), and the LDM failed to map effectively to this new adversarial latent space (figure 4.17).



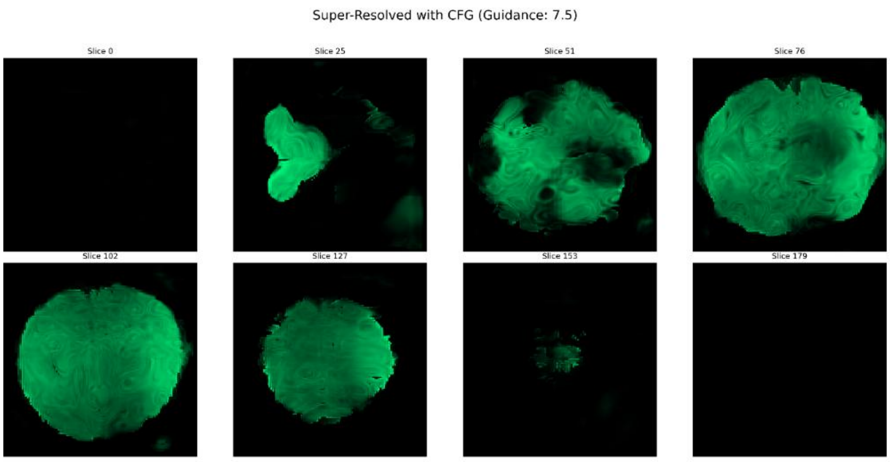
4.16. Figure: The second adversarial autoencoder reconstruction without skip connections (huge visual improvement)



4.17. Figure: SR with the second adversarial finetuning attempt

4.5.4 Final Model Performance

The final iteration involved fine-tuning the diffusion model to align with the adversarially trained decoder. This achieved a PSNR of 22.69 dB. While this remains quantitatively lower than the bicubic baseline this is a known phenomenon in generative super-resolution (the perception-distortion trade-off). Bicubic interpolation minimizes MSE by averaging pixels, leading to high PSNR but blurry textures. The LDM prioritizes perceptual sharpness, which mathematically deviates from the ground truth (lowering PSNR) but recovers the "appearance" of realistic anatomy, as seen in the preservation of edge structures in Figure 4.18. The model successfully synthesizes plausible anatomical structures, serving as a proof-of-concept for LDM-based MRI upscaling.



4.18. Figure: Super-resolved image with the second adversarial autoencoder, with a guidance factor of 7.5 and using 1000 diffusion steps.

Model Variant	PSNR (dB)	SSIM	PSNR (vs. Baseline)	Mean Intensity Shift	Qualitative Observations
Bicubic Baseline	**27.73**	**0.84**	—	—	Smooth, lacks fine detail
Initial LDM	18.87	0.49	-8.86	27.18%	Severe artifacts, poor structure (see **Fig. 4.13**)
CFG Fine-Tuned LDM	23.56	0.65	-4.17	0.12%	Better alignment, visible improvement (see **Fig. 4.14**)
1st Adversarial Attempt	11.17	0.22	-16.56	107.64%	Unstable, unusable output (see **Fig. 4.15**)
2nd Adversarial (Anchored)	12.26	0.28	-15.47	368.05%	Sharper but artifact-heavy (see **Fig. 4.17**)
Final Aligned Model	**22.69**	**0.61**	**-5.04**	**3.12%**	**Coherent structure, proof-of-concept (see Fig. 4.18)**

4.1. Table: Quantitative Summary of Key Experiments

4.5.5 Summary of Findings

The evaluation reveals a clear trajectory:

- The VQ-VAE is a powerful compressor but suffers from a critical dependency on skip connections.
- The initial LDM failed primarily due to latent space misalignment.
- Classifier-free guidance (CFG) proved to be the pivotal correction for this misalignment. By enforcing stricter adherence to conditioning, CFG indirectly regularized the latent statistics, significantly recovering performance.
- Adversarial training enhanced sharpness but destabilized latent-space consistency.
- The final aligned model achieved coherent reconstructions below bicubic numerically but with superior perceptual quality, confirming the potential of LDM-based fMRI enhancement.

5 Overall Conclusions and Future Work

This research investigated the application of generative artificial intelligence to functional magnetic resonance imaging (fMRI), progressing from synthetic volume generation to the clinical task of super-resolution. The work culminated in the development of a latent diffusion model (LDM) framework that integrates VQ-GAN autoencoding with conditional diffusion processes. Through systematic iteration of fine-tuning strategies, including classifier-free guidance and adversarial decoding, the study describes the specific architectural constraints for diffusion-based 3D medical imaging.

5.1 Conclusion

The main outcome of this thesis is the successful design and evaluation of a two-stage LDM pipeline for fMRI image enhancement. Several key findings emerge:

- **Synthetic image generation as a foundation**: Early attempts at unconditional generation highlighted the limitations of small and 2D latent spaces and weak conditioning, but they also established the feasibility of using structured latent manifolds for brain-like reconstructions. These experiments informed the architectural and methodological choices that enabled later success in super-resolution.

- **Latent space structure is crucial**: The transition from a standard autoencoder to a VQ-GAN was essential. Its discrete, codebook-based latent space provided the stability needed for diffusion modeling, mitigating the "latent mismatch" problem observed in initial experiments, future work suggests a larger codebook is required for optimal reconstruction.

- **Classifier-free guidance (CFG) as the most robust strategy**: CFG proved to be the most effective improvement. Beyond simply following the input condition, the retraining required for CFG aligned the generated latents with the real data, enhancing performance.

- **Decoder dependency remains the bottleneck**: The decoder's reliance on skip connections, while advantageous during training, collapsed performance during generative inference. This "no-skip" limitation ultimately capped quantitative performance, even when perceptual sharpness improved.

- **Adversarial fine-tuning adds sharpness but instability**: Introducing adversarial training improved perceptual realism and reduced skip reliance, but it destabilized latent alignment unless carefully regularized. While fragile, these experiments demonstrated that texture and sharpness gains are possible if balanced against structural fidelity.

Overall, while the models did not outperform bicubic interpolation on PSNR or SSIM, they consistently produced sharper and more coherent reconstructions. This confirms that diffusion models, when paired with a stable, discrete latent space, form a viable basis for future fMRI super-resolution systems.

5.2 Future Works

Building on these findings, future research should aim to overcome the decoder bottleneck and refine end-to-end compatibility between the autoencoder and diffusion model:

- **End-to-end harmonization**: The most promising direction is joint finetuning of the autoencoder and diffusion model, ideally combining stabilized adversarial fine-tuning for robust no-skip reconstruction with retraining the diffusion model on these updated latents.

- **Advanced decoder architectures**: Future work should explore decoders less dependent on skip connections, possibly through richer latent encodings (e.g. bigger codebook) or alternative mechanisms for transmitting spatial information. This could resolve the fundamental generative limitation observed here.

- **Perceptual and task-specific metrics**: Traditional metrics such as PSNR and SSIM undervalued perceptual gains. Incorporating perceptual metrics (LPIPS, DISTS) and downstream clinical tasks (e.g., lesion detection) into evaluation will provide a more accurate measure of practical impact.

- **Scaling and clinical validation**: Extending the pipeline to other datasets and, validating with the help of radiologists, would be essential steps toward a trustworthy system.

By addressing these directions, diffusion-based pipelines can advance from proof-of-concept toward clinically ready systems, offering faster acquisitions, improved resolution of legacy datasets, and ultimately more effective diagnostic workflows in functional neuroimaging.

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